

ECEN-635: Homework 2

Due on September 10, 2025 at 4:30 pm

Instructor: Dr. Robert Nevels

Brody Jordan
431005357

Problem 1

Convert the following time domain expressions for the electric and magnetic field into frequency and domain expressions. Show each mathematical step in your calculations. A miraculous answer without the complete calculation will be judged harshly.

- (a) $\mathbf{E}(t) = 4 \cos(\omega t - 28z) \hat{\mathbf{y}}$ V/m
- (b) $\mathbf{E}(t) = 9 \sin(\omega t - 23z) \hat{\mathbf{x}}$ V/m
- (c) $\mathbf{H}(t) = 7 \cos(\omega t - 23z) \hat{\mathbf{x}}$ mA/m
- (d) $\mathbf{E}(t) = 4.6 e^{-48\pi z} \cos(\omega t - 139z + \pi/4) \hat{\mathbf{y}}$ V/m

Solution

Problem 2

Convert the following frequency domain expressions to the time domain. Show each mathematical step in your calculation.

(a) $\mathbf{E} = 15e^{-j23z}\hat{\mathbf{x}}$ V/m

(b) $\mathbf{E} = 32e^{-(52+j135)z}\hat{\mathbf{y}}$ V/m

(c) $\mathbf{H} = [(2+j2)\hat{\mathbf{x}} + (1-j3)\hat{\mathbf{y}}]e^{-j245z}$ A/m

(d) $\rho = 3je^{j\pi/2}$ C/m³

Be sure to place each of the vector components in simplest magnitude and phase form before converting to the time domain.

Solution

Problem 3

Assuming a homogeneous region where there is no current, charge, or loss:

- (a) Use Maxwell's equations to convert the frequency domain electric field expression in 2(b) to the corresponding magnetic field. Show all mathematical steps.
- (b) Use Maxwell's equations to convert the frequency domain magnetic field in 2(c) into a frequency domain electric field into a frequency domain magnetic field. Show all mathematical steps.

Solution

Problem 4

Use 1(b) and the corresponding electric field found in problem 1(c) to calculate the time average power density. Show all mathematical steps.

Solution

Problem 5

Starting with the frequency domain form of Maxwell's equations, derive the magnetic field vector Helmholtz equation. Number your Maxwell equations and at every step write briefly which Maxwell equation and/or which identity you are using from the Vector Identities section of Appendix II.3.1 (pg. 958). Assume there is no electric or magnetic current and no electric or magnetic charge.

Solution